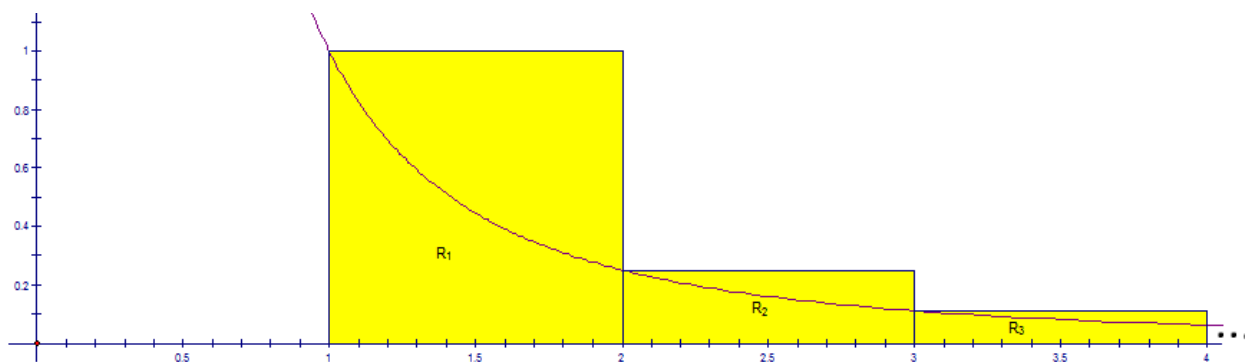


3.4.1 The Integral Test

Motivation: Finding the exact sum of a series is, in general, difficult. This is because the formula for the n^{th} partial sum of a series is usually quite complicated. Due to this, we will concentrate in the next few sections on simply determining whether or not a series converges without explicitly finding the sum. We will also discuss some ways to obtain estimates for the sum of convergent series.

One series which does not have a “nice” formula for its n^{th} partial sum is: $\sum_{k=1}^{\infty} \frac{1}{k^2}$. Suppose we want to know whether this series converges or diverges.

In order to attack this question we will consider the integral: $\int_1^{\infty} \frac{1}{x^2} dx$. Suppose we approximate this integral (with rectangles) by partitioning the interval $[1, \infty)$ into subintervals of length 1 and use left endpoints of the subintervals to establish the height of our rectangles. A picture of this is below:



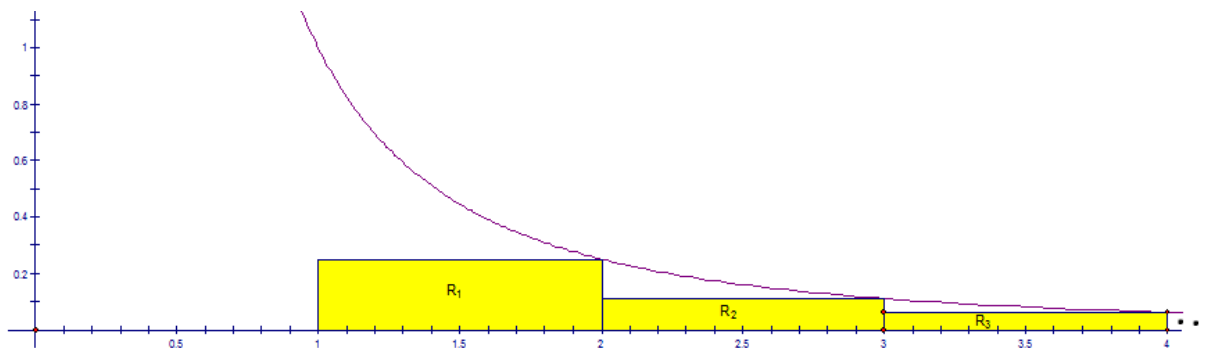
We see that the area of R_1 is 1, the area of R_2 is $\frac{1}{4}$, and the area of R_3 is $\frac{1}{9}$. In general, there will be an infinite number of rectangles drawn under the curve and the area of R_i will be $\left(\frac{1}{i}\right)^2$. A formula for the sum of the area of all the rectangles under the curve would be $\sum_{k=1}^{\infty} \frac{1}{k^2}$.

Since this approximation is clearly an overestimate it is reasonable to conclude that if $\int_1^{\infty} \frac{1}{x^2} dx$ is divergent, then $\sum_{k=1}^{\infty} \frac{1}{k^2}$ is divergent.

We note that $\int_1^{\infty} \frac{1}{x^2} dx = \lim_{t \rightarrow \infty} \int_1^t \frac{1}{x^2} dx = \lim_{t \rightarrow \infty} \left(\frac{-1}{t} + 1 \right) = 1$. Therefore, we cannot conclude that $\sum_{k=1}^{\infty} \frac{1}{k^2}$ diverges.

Note: In fact in higher level mathematics it is possible to prove that $\sum_{k=1}^{\infty} \frac{1}{k^2} = \frac{\pi^2}{6}$.

On the other hand, suppose we approximate this integral (with rectangles) by partitioning the interval $[1, \infty)$ into subintervals of length 1 and use right endpoints of the subintervals to establish the height of our rectangles. A picture of this is below:



We see that the area of R_1 is $\frac{1}{4}$, the area of R_2 is $\frac{1}{9}$, and the area of R_3 is $\frac{1}{16}$. In general, there will be an infinite number of rectangles drawn under the curve and the area of R_i will be $\left(\frac{1}{i+1}\right)^2$. A formula for the sum of the area of all the rectangles under the curve would be $\sum_{i=1}^{\infty} \frac{1}{(i+1)^2} = \sum_{k=2}^{\infty} \frac{1}{k^2}$.

Clearly this approximation is an underestimate of $\int_1^{\infty} \frac{1}{x^2} dx$. So, if $\int_1^{\infty} \frac{1}{x^2} dx$ is convergent, $1 + \int_1^{\infty} \frac{1}{x^2} dx$ will bound the partial sums of $\sum_{k=1}^{\infty} \frac{1}{k^2}$ from above, and we will have that $\sum_{k=1}^{\infty} \frac{1}{k^2}$ is convergent since the sequence of its partial sums is increasing (by the monotonic sequence theorem).

Note: The above motivating example leads to the main theorem of the section.

Theorem (The Integral Test): Suppose f is a continuous, positive, decreasing function on $[1, \infty)$ and let $a_n = f(n)$. Then,

1. If $\int_1^{\infty} f(x) dx$ is convergent, then $\sum_{n=1}^{\infty} a_n$ is convergent.
2. If $\int_1^{\infty} f(x) dx$ is divergent, then $\sum_{n=1}^{\infty} a_n$ is divergent.

Proof: See the textbook.

Note: More generally, the integral test will work regardless of where the index begins. So if the index begins at “a” and we have a decreasing positive function on $[a, \infty)$, we can apply the integral test.

Example 1: Determine (with justification) whether the series $\sum_{k=1}^{\infty} \frac{10}{1+k^2}$ converges or diverges.

First we note that the function $f(x) = \frac{10}{1+x^2}$ is continuous, positive, and decreasing on $[1, \infty)$. Now, we calculate that:

$$\begin{aligned} \int_1^{\infty} \frac{10}{1+x^2} dx &= \lim_{t \rightarrow \infty} \int_1^t \frac{10}{1+x^2} dx = \lim_{t \rightarrow \infty} \left[10 \tan^{-1}(x) \Big|_1^t \right] = \lim_{t \rightarrow \infty} [10 \tan^{-1}(t) - 10 \tan^{-1}(1)] \\ &= 10 \left(\frac{\pi}{2} \right) - 10 \left(\frac{\pi}{4} \right) = \frac{5\pi}{2} \end{aligned}$$

Since $\int_1^{\infty} \frac{10}{1+x^2} dx$ is convergent, we know that $\sum_{k=1}^{\infty} \frac{10}{1+k^2}$ is convergent by the integral test. ■

Example 2: What is the simplest way to determine whether the series $\sum_{k=1}^{\infty} \frac{k+2}{k+1}$ converges or diverges?

Here it is easy to answer the question by looking at the divergence test. We let $f(x) = \frac{x+2}{x+1}$ and note that

$\lim_{x \rightarrow \infty} f(x) = 1$. Therefore $\lim_{k \rightarrow \infty} \frac{k+2}{k+1} = 1$ and by the divergence test the series $\sum_{k=1}^{\infty} \frac{k+2}{k+1}$ diverges. ■

Alternatively, we could have shown the series diverges using the integral test (details below).

Let $f(x) = \frac{x+2}{x+1}$. Note that $f(x)$ can be simplified as follows: $f(x) = \frac{x+1+1}{x+1} = \frac{x+1}{x+1} + \frac{1}{x+1} = 1 + \frac{1}{x+1}$.

From here we can see that $f(x)$ is a continuous, positive, decreasing function on $[1, \infty)$, so the integral test applies.

$$\int_1^{\infty} \left(1 + \frac{1}{x+1} \right) dx = \lim_{t \rightarrow \infty} \left(\int_1^t \left(1 + \frac{1}{x+1} \right) dx \right) = \lim_{t \rightarrow \infty} \left((x + \ln |x+1|) \Big|_1^t \right) = \lim_{t \rightarrow \infty} (t + \ln |t+1| - 1 - \ln |2|) = \infty$$

Therefore, by the integral test, since $\int_1^{\infty} \left(1 + \frac{1}{x+1} \right) dx$ diverges the series $\sum_{k=1}^{\infty} \frac{k+2}{k+1}$ diverges. ■